

1. A particle P has mass $3m$ and a particle Q has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately before the collision the speed of P is ku , where k is a constant, and the speed of Q is $2u$.

Immediately after the collision the speed of P is u and the speed of Q is $3u$.

The direction of motion of Q is reversed by the collision.

- (a) Find, in terms of m and u , the magnitude of the impulse exerted on Q by P in the collision.

(2)

- (b) Find the two possible values of k .

(5)

2. A car moves along a straight horizontal road with constant acceleration $a \text{ m s}^{-2}$ where $a > 0$

The car is modelled as a particle.

At time $t = 0$, the car passes point A and is moving with speed $u \text{ m s}^{-1}$

In the first three seconds after passing A the car travels 20 m.

In the fourth second after passing A the car travels 10 m.

The speed of the car as it passes point B is 20 m s^{-1}

Find the time taken for the car to travel from A to B .

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, are given by

$$\mathbf{F}_1 = (5\mathbf{i} + 2\mathbf{j})\text{N} \quad \mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})\text{N} \quad \mathbf{F}_3 = (a\mathbf{i} + b\mathbf{j})\text{N}$$

where a and b are constants.

The forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle P of mass 4 kg.

Given that P rests in equilibrium on a smooth horizontal surface under the action of these three forces,

- (a) find the size of the angle between the direction of $\mathbf{F}_3$ and the direction of $-\mathbf{j}$. (4)

The force $\mathbf{F}_3$ is now removed and replaced by the force $\mathbf{F}_4$ given by $\mathbf{F}_4 = \lambda(\mathbf{i} + 3\mathbf{j})\text{N}$, where λ is a positive constant.

When the three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_4$ act on P , the acceleration of P has magnitude 3.25 m s^{-2}

- (b) Find the value of λ . (5)
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4.

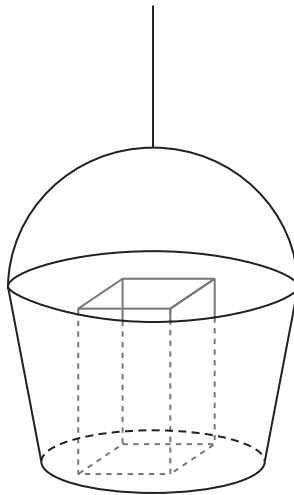


Figure 1

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

When the bucket is on the ground, a bag of cement of mass 25 kg is placed in the bucket.

The bucket with the bag of cement moves vertically upwards with constant acceleration 0.2 m s^{-2} . Air resistance is modelled as being negligible.

(a) Find the tension in the cable.

(3)

At the top of the building, the bag of cement is removed. A box of tools of mass 12 kg is now placed in the bucket.

Later on the bucket with the box of tools is moving vertically downwards with constant deceleration 0.1 m s^{-2} . Air resistance is again modelled as being negligible.

(b) Find the magnitude of the normal reaction between the bucket and the box of tools.

(3)

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

A particle P is moving with constant acceleration. At 2pm, the velocity of P is $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$ and at 2.30pm the velocity of P is $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time T hours after 2pm, P is moving in the direction of the vector $(-\mathbf{i} + 2\mathbf{j})$

(a) Find the value of T .

(6)

Another particle, Q , has velocity $\mathbf{v}_Q \text{ km h}^{-1}$ at time t hours after 2pm, where

$$\mathbf{v}_Q = (-4 - 2t)\mathbf{i} + (\mu + 3t)\mathbf{j}$$

and μ is a constant.

Given that there is an instant when the velocity of P is equal to the velocity of Q ,

(b) find the value of μ .

(3)

6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane.
- (3)

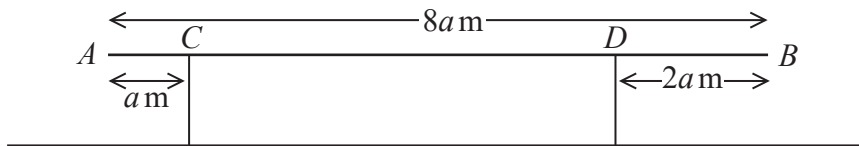
The particle comes to instantaneous rest at the point B .

- (b) Find the distance AB .
- (5)

The particle now slides down the plane from B . At the instant when the particle passes through the point C on the plane, the speed of the particle is again 5 m s^{-1}

- (c) Find the distance BC .
- (5)
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7.

**Figure 2**

A non-uniform beam AB , of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C , where $AC = a$ metres and the other support is at D , where $DB = 2a$ metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at D is three times the magnitude of the normal reaction between the beam and the support at C .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance x metres from A ,

(a) find an expression for x in terms of a .

(5)

A box of mass $M \text{ kg}$ is placed on the beam at E , where $AE = 2a$ metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at C is now equal to the magnitude of the normal reaction between the beam and the support at D .

By modelling the box as a particle,

(b) find the value of M .

(5)

8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram A and a speed-time graph for the motion of tram B , from $t = 0$ to the instant when tram B overtakes tram A .

(3)

At the instant when the two trams are moving with the same speed, tram A is d metres in front of tram B .

- (b) Find the value of d .

(5)

- (c) Find the distance of the trams from the depot at the instant when tram B overtakes tram A .

(5)
